

ELECTRICAL SAFETY TOOLBOX TALK

Control electric shock, arc-flash and fire hazards.

8-10 MINUTES

OBJECTIVE

Review electric-shock, arc-flash, burn, fire and unexpected-energisation hazards associated with electrical equipment, temporary power, cables, sockets, panels and energy sources, and verify the required controls before work begins.

TOPIC EXPLANATION

- Electricity is an invisible hazard. A damaged cable, loose connection, open panel, poor earthing or wet environment can expose a worker to a fatal electric current.
- Electrical equipment must only be installed, repaired or opened by authorised personnel. Unauthorised workers must not remove covers, replace protective devices or work on energised circuits.
- De-energised work must be the normal approach. Before intervention, power must be isolated, LOTO applied, absence of voltage verified with a suitable tester and re-energisation prevented.

REALISTIC SITE SCENARIO

- A worker used a portable electric tool without inspecting the supply cable. The outer insulation had been cut and the conductor was exposed.
- The tool was used in a damp area and the worker contacted the damaged section, receiving an electric shock. Suitable residual-current protection was not installed.
- The incident could have been prevented through pre-use inspection, quarantine of damaged equipment, suitable RCD/GFCI protection and dry working conditions.

REMEMBER

Do not use electrical equipment showing damage, exposed conductors, burning smell or abnormal heat. Isolate the supply, tag the equipment and report it to an authorised person.

ELECTRICAL SAFETY TOOLBOX TALK

Hazards, control measures and daily pre-work verification.

8-10 MINUTES

KEY HAZARDS

- Damaged cables, plugs, sockets or equipment enclosures.
- Open or unlocked electrical panels.
- Unsuitable equipment used in wet conditions.
- Poor earthing or missing residual-current protection.
- Overloaded extension leads and multiple adapters.
- Unauthorised work on energised circuits.
- Arc flash, short circuits and hot surfaces.
- Cables damaged in access routes or creating trip hazards.

CONTROL MEASURES

- Inspect equipment, cables, plugs and sockets before use.
- Remove damaged equipment from service and quarantine it.
- Verify suitable earthing and residual-current protection.
- Keep electrical panels closed, locked and accessible.
- Protect cables from water, sharp edges and vehicle routes.
- Do not overload or daisy-chain extension leads.
- Prioritise de-energised work and apply LOTO.
- Verify absence of voltage with a suitable tester.
- Allow only authorised electrical personnel to intervene.
- Safely isolate power during electrical faults or fires.

SUPERVISOR TALKING SCRIPT

Team, before starting today, we will inspect all cables, sockets and electrical equipment. We will not use anything damaged, wet, showing exposed conductors or abnormal heat. Only authorised personnel may access electrical panels. Before intervention, power will be isolated, LOTO applied and absence of voltage verified.

DISCUSSION QUESTIONS

- Have all cables and electrical tools been inspected?
- Are earthing and residual-current protection suitable?
- Are cables protected from water, edges and vehicles?
- Are electrical panels closed and locked?
- What action will be taken if damaged equipment is found?
- Who will complete the isolation and electrical testing?

VERIFY BEFORE STARTING TODAY

- | | |
|---|---|
| ☐ Cables intact | ☐ Plugs intact |
| ☐ Sockets suitable | ☐ Earthing available |
| ☐ RCD/GFCI active | ☐ Panels closed |
| ☐ Area dry | ☐ Cables protected |
| ☐ LOTO ready | ☐ Authorised person identified |

ATTENDANCE AND APPROVAL

ELECTRICAL SAFETY TOOLBOX TALK

Project / Site

Date

Presented by

Work area

No	Full Name	Company / Role	Signature
1			
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Supervisor notes / additional site instructions